



Sprint 0 Document

Massar مسار



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Introduction to Massar

What is Massar?

Massar is a smart, **adaptive learning** platform designed to help students master new subjects efficiently. Instead of following a "one-size-fits-all" plan, Massar uses AI to create a personalized path for every learner. By breaking down complex PDFs and slideshows into small, manageable pieces, the system generates adaptive quizzes that ensures you truly understand the basics before moving on to harder topics.

Why Use It?

Most online learning is just reading. Massar changes this by:

- **Analyzing your progress:** It tracks what you know and what you don't.
- **Guiding your path:** It requires 95% mastery of a topic before unlocking the next level.
- **Providing AI support:** It generates custom study guides and mind maps exactly when you need them.

Stakeholders

To ensure the platform runs smoothly, several groups are involved:

- **Students:** The primary users who use Massar to organize their studies, track their growth, and achieve mastery.
- **Administrators:** The staff who manage user accounts, monitor platform health, and view overall learning statistics.



Technical Team: The developers and AI specialists responsible for maintaining the "BKT" engine and ensuring data security.

Glossary

Term	Definition
Adaptive Learning	A delivery method where the system changes the content and difficulty in real-time based on the learner's individual performance.
Bayesian Knowledge Tracing (BKT)	An algorithm used to model a learner's mastery of a specific skill over time by calculating the probability that they know a concept based on their quiz answers.
Knowledge Component (KC)	The smallest "atomic" unit of information (a fact, rule, or concept) that can be learned and tested independently.
Semantic Deconstruction	The process where the AI "reads" a PDF or PPT to identify key concepts and the logical relationships between them.



Mastery Threshold	The specific success rate (set at 95% for Massar) a student must reach in a quiz before the system considers the topic "learned" and unlocks new content.
Learning Command Center	The primary user interface for students that centralizes subject organization, mastery tracking, and progress visualization.

Product Backlog/Requirement

#	Functional Requirements (Features)	Actor
1	The system shall allow users to create a new account (Sign Up) by providing their name, email, and password.	User
2	The system shall allow users to log in to the platform using their registered email and password.	User
3	The system shall implement Role-Based Access Control (RBAC) to redirect users to their specific dashboard: Admin, Student, or Instructor.	User
4	The system shall allow users to manage their personal profiles (Name, email, password).	User
5	The system shall provide a centralized dashboard for students to organize subjects into specific categories.	Student
6	The system shall allow students to track overall mastery for each subject based on aggregated KC data.	Student
7	The system shall provide a visual timeline of the student's cognitive growth over time.	Student
8	The system shall allow students to generate a custom Learning Module by defining a topic name, which the system populates with relevant Knowledge Components (KCs).	Student

9	The system shall allow students to upload educational resources in PDF and PPT formats.	Student
10	The system shall deconstruct uploaded files into granular KCs using AI-driven semantic analysis.	System
11	The system shall allow students to manually edit or delete AI-generated KCs to ensure 100% accuracy.	Student
12	The system shall utilize a Bayesian Knowledge Tracing (BKT) engine to update mastery probability (0.0 to 1.0) after every user response.	System
13	The system shall generate adaptive quizzes where question difficulty is determined by the BKT engine state.	Student
14	The system shall provide a visual performance breakdown after each quiz, highlighting "Focus Areas".	System
15	The system shall allow students to request AI-generated Mind Maps and Summary Sheets for unmastered topics.	Student
16	The system shall enforce a 95% mastery threshold , locking advanced content until proficiency is achieved.	System
17	The system shall persist all mastery states and resources in a secure database (Firebase) for cross-device access.	System
18	The system shall allow Admin to view, edit, and delete user accounts (Student/Admin/Instructor).	Admin

19	The system shall allow Admin to view a dashboard showing platform-wide engagement statistics.	Admin
20	The system shall allow users to securely log out and terminate the active session.	User
21	The system shall allow Instructors to create groups and add students via email or unique IDs.	Instructor
22	The system shall allow Instructors to upload specific PDF/PPT content for their groups to generate custom quizzes.	Instructor
23	The system shall provide a dashboard to monitor real-time group progress and average mastery levels.	Instructor
24	The system shall identify the specific strengths ($SP(L) > 0.90$) and weaknesses for each student.	Instructor
25	The system shall allow Instructors to initiate and participate in a group chat with all enrolled students.	Instructor
26	The system shall allow Instructors to engage in private, one-on-one messaging with individual students.	Instructor
27	The system shall allow Students to participate in group discussions with peers and the instructor.	Student
28	The system shall allow Students to send private help requests directly to their instructor.	Student

User Stories

#	Title	Description	Story Points	Priority	Actor
1	Account Creation	As a new user, I want to create a new account by providing my name, email, and password, so that I can access the platform's features.	2	Must-have	User
2	Secure Login	As a registered user, I want to log in to the platform, so that I can securely access my personalized learning data.	1	Must-have	User
3	Profile Management	As a user, I want to manage my personal profile (name, email, and password), so that my account information stays accurate and secure.	2	Should-have	User
4	Learning Command Center	As a student, I want a centralized "Learning Command Center," so that I can organize my subjects and track my cognitive growth in one place.	5	Must-have	Student
5	Mastery Tracking	As a student, I want to see a Mastery Percentage for each subject in the Command Center, so that I know exactly how much of the course I have learned.	3	Must-have	Student
6	Cognitive Growth Timeline	As a student, I want to view a visual timeline of my cognitive growth, so that I	3	Should-have	Student

		can track my learning progress milestones over time.			
7	Module Generation	As a student, I want to generate custom Learning Modules by defining a topic name, so that the system can automatically populate it with KCs.	3	Must-have	Student
8	Resource Upload	As a student, I want to upload educational resources in PDF and PPT formats, so that I can use my own study materials as a basis for learning.	3	Must-have	Student
9	AI Deconstruction	As a student, I want the system to deconstruct my uploaded files into granular KCs using AI, so that I can study in smaller, manageable steps.	8	Must-have	System
10	KC Refinement	As a student, I want to manually edit or delete the AI-generated KCs, so that I can ensure the learning structure is 100% accurate.	3	Should-have	Student
11	Adaptive Quizzes	As a student, I want to take quizzes where the difficulty is determined by the BKT engine, so that I am always challenged appropriately.	8	Must-have	Student
12	BKT Real-time Updates	As a student, I want the system to update my mastery probability (0.0 to 1.0) after every response, so that my path adapts in real-time.	5	Must-have	System
13	Performance Visuals	As a student, I want to view a visual breakdown of my performance after each quiz, so that I can clearly identify my "Focus Areas".	3	Should-have	System

14	AI Study Aids	As a student, I want to request AI-generated Mind Maps and Summary Sheets, so that I can get extra help with difficult topics.	5	Could-have	Student
15	Mastery Gatekeeping	As a student, I want the system to enforce a 95% mastery threshold, so that I build a strong foundation before unlocking advanced content.	3	Must-have	System
16	Instructor Group Creation	As an instructor, I want to create groups and add my students, so that I can manage my class cohorts and provide specific curriculum.	3	Must-have	Instructor
17	Group Performance View	As an instructor, I want to view a performance dashboard for my whole group, so that I can monitor the real-time progress of my entire class.	5	Must-have	Instructor
18	Strength & Weakness Analysis	As an instructor, I want to see the specific strengths and weaknesses of each student, so that I can understand where individual learners need extra help.	5	Should-have	Instructor
19	Group Communication	As an instructor, I want to chat with my whole group, so that I can send announcements and answer general questions for all students at once.	3	Should-have	Instructor
20	Private Mentorship	As an instructor, I want to start a private chat with a student, so that I can provide one-on-one guidance and discuss their specific progress.	3	Should-have	Instructor
21	Student Group Chat	As a student, I want to participate in group discussions, so that I can collaborate with	3	Should-have	Student

		my classmates and learn from their questions.			
22	Private Help Requests	As a student, I want to message my instructor privately, so that I can ask confidential questions about my performance or the course material.	3	Should-have	Student
23	Data Persistence	As a user, I want my mastery states and resources persisted in a secure database, so that I can resume learning across any device.	5	Must-have	System
24	User Administration	As an admin, I want to view, edit, and delete user accounts, so that I can manage the platform's user base and resolve issues.	3	Should-have	Admin
25	Platform Analytics	As an admin, I want to view a dashboard showing platform-wide statistics, so that I can monitor overall engagement and trends.	3	Could-have	Admin
26	Secure Logout	As a user, I want to log out of my account, so that my session is terminated and my data remains secure on shared devices.	1	Must-have	User

Non-Functional Requirements

Category	Requirement Description
Security	The system must implement robust authentication and authorization protocols to ensure that users can only access their authorized data and dashboards.
Security	All communication between users, including group and private chats, must be encrypted to ensure data privacy and prevent unauthorized access.
Performance	The automated deconstruction engine should be optimized to process standard educational documents (PDF/PPT) within a time frame that ensures a smooth user experience.
Performance	The messaging system must support real-time interaction with minimal latency to facilitate seamless communication between instructors and students.
Scalability	The system architecture must be designed to handle a high volume of concurrent users and data processing requests without a loss in performance or stability.
Reliability	The mastery tracking engine must ensure data integrity, accurately recording and saving all progress and quiz results after every user interaction.
Availability	The platform should maintain high uptime and accessibility to allow students and instructors to use the system whenever needed across various devices.
Usability	The user interface must be fully responsive and optimized for a variety of screen sizes, including desktop monitors and mobile tablets.
Usability	The platform must provide clear, intuitive visual indicators of progress and performance to help users navigate their learning paths effectively.

Infrastructure Readiness

#	Type	Name	Status
1	Software	Discord	Available
2	Software	Trello	Available
3	Hardware	PC	Available
4	Network	Wi-Fi	Available
5	Software	WhatsApp	Available

Environment Specification

#	Type	Name	Status
1	Software	VScode	Available
2	Software	Github	Available
3	Database tool	Firebase	Available
4	Programming Language	Python	Available
5	JavaScript Library	React	Available

Team Readiness

Role	Responsibilities	Assigned To
Backend Developer	Build APIs using FastAPI, implement BKT logic, manage Fairbase	Backend Team

Frontend Developer	Build UI using React, integrate with API, manage state, implement quiz interface	Frontend Team
AI Integration	Handle quiz generation logic, content processing from PDFs, ensure question validation	AI Member
Project Manager	Sprint planning, backlog management, documentation	PM

Training

ID	Role	Training Topic	Description
T1	Backend	FastAPI Structure	Learn routers, services, schemas, dependency injection
T2	Backend	Fairbase	Database schema design & relationships
T3	Frontend	React Basics	Components, hooks, API calls
T4	All	gitHub	control software versions and manage source code changes effectively

Knowledge Acquisition (Domain)

ID	Knowledge Area	Description	Priority
K1	Adaptive Learning	Understand mastery threshold logic	Must
K2	Knowledge Components (KCs)	How to model skills per subject	Must
K3	AI-generated Questions	Validation & fallback mechanism	Should
K4	REST API Design	Status codes & versioning best practices	Must



Design/Architecture:

Architecture Overview:

We chose a **multi-tier architecture** for **Massar** because it makes the system more secure, organized, and easy to scale. Each layer has a clear role, so everything is separated and easier to manage. It also protects sensitive data since the database is never directly exposed. This way, we can improve or scale one part without affecting the whole system.

The system is divided into three tiers:

Presentation Tier (Client-Side):

Technologies: React.js, Tailwind CSS.

Responsibility: This tier handles everything the user sees and interacts with, like quizzes and file uploads. It collects input and sends it to the server.

Functions:

Displays the user interface (quizzes, dashboards, uploads).

Collect and validate user input.

Sends requests to the server and shows responses.

- Application Tier (Server-Side):

Technologies: Node.js (Express), Python (Fast API).

Responsibility: This is where the main logic runs, like calculating learning progress, applying rules, and connecting to AI. It processes requests and controls how data moves.

Functions:

Processes business logic and system rules.

Communicates with the database and external APIs.



Handles authentication and access control.

- **Data Tier (Persistence):**

Technologies: Firebase.

Responsibility: This tier stores all the data safely, including user records and files. It makes sure information is saved and not lost.

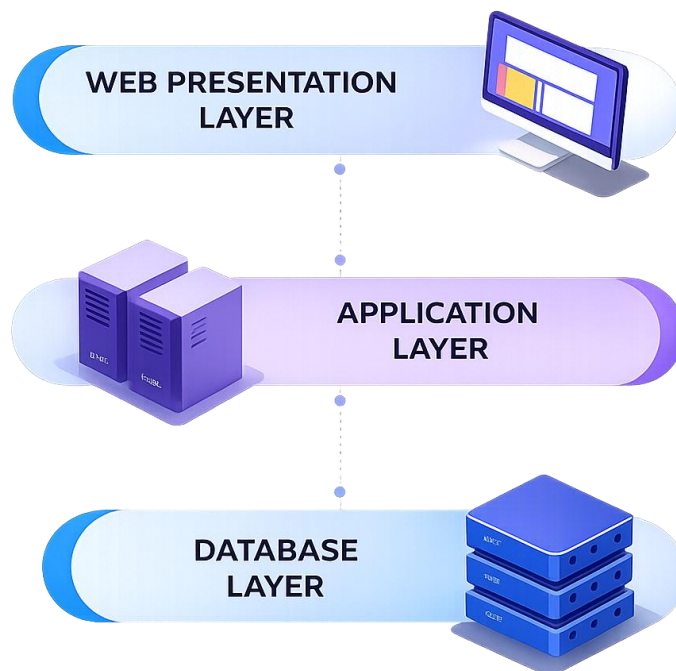
Functions:

Stores user data, results, and files.

Manages data retrieval and updates.

Ensures data security and persistence.

Multi-Tier Architecture



Project Release Road-map

#	Feature	Sprint	Story Points	Priority
Sprint 1: Basic Account Management and UI				
1	Account Creation	Sprint 1	2	Must-have
2	Secure Login	Sprint 1	1	Must-have
3	Profile Management	Sprint 1	2	Should-have
4	Learning Command Center	Sprint 1	5	Must-have
5	Mastery Tracking	Sprint 1	3	Must-have
6	Cognitive Growth Timeline	Sprint 1	3	Should-have
7	User Administration	Sprint 1	3	Should-have
8	Platform Analytics	Sprint 1	3	Could-have
9	Secure Logout	Sprint 1	1	Must-have

Sprint 2: AI Core and Content Ingestion				
10	Module Generation	Sprint 2	3	Must-have
11	Resource Upload	Sprint 2	3	Must-have
12	AI Deconstruction	Sprint 2	8	Must-have
13	KC Refinement	Sprint 2	3	Should-have
Sprint 3: Adaptive Learning and Assessment				
14	Adaptive Quizzes	Sprint 3	8	Must-have
15	BKT Real-time Updates	Sprint 3	5	Must-have
16	Performance Visuals	Sprint 3	3	Should-have
17	AI Study Aids	Sprint 3	5	Could-have
18	Mastery Gatekeeping	Sprint 3	3	Must-have
Sprint 4: Collaboration and Data Persistence				

19	Instructor Group Creation	Sprint 4	3	Must-have
20	Group Performance View	Sprint 4	5	Must-have
21	Strength & Weakness Analysis	Sprint 4	5	Should- have
22	Private Mentorship	Sprint 4	3	Should- have
23	Student Group Chat	Sprint 4	3	Should- have
24	Data Persistence	Sprint 4	5	Must-have

Project Management Plan (High Level)

- In Scope:

- User Management and Authentication: *User registration (Sign Up) with email and password, Secure login and logout functionality, Role based access control for Student, Instructor, and Admin.*
- Learning Command Center dashboard For Student: *Centralized dashboard for students, Subject organization into categories.*
- AI Learning Support Features: *AI generated summary sheets, Strength and weakness detection per Knowledge Component.*
- Adaptive Quiz Engine (BKT): *Quiz generation based on defined Knowledge Components, Adaptive question selection based on student performance, Visual performance breakdown after each quiz.*
- Instructor Features: *Instructor account creation, Real time group performance monitoring dashboard, Individual student analytics.*
- Admin Features: *View, edit, and delete user accounts, manage user roles, Monitor system wide statistics.*

- Out Scope:

- Native mobile applications.
- Third-party LMS integrations.

- **Assumptions:**

- *It is assumed that the selected technology stack (React, FastAPI, Firebase, and BKT engine) will meet the project's performance and scalability requirements.*
- *It is assumed that the AI based semantic analysis will generate knowledge components with acceptable accuracy.*
- *It is assumed that students and instructors will have reliable internet access and basic digital literacy.*
- *It is assumed that students will actively use the system by uploading materials, completing quizzes, and reviewing feedback to ensure accurate progress tracking and mastery evaluation.*

- **Dependencies:**

- *The project depends on the successful implementation of the BKT engine after finalizing the knowledge component structure.*
- *The project depends on stable and fully developed backend APIs to enable proper frontend integration.*
- *The project depends on Trello for task management and progress tracking, and on Discord and WhatsApp for efficient team communication.*
- *The project depends on successful completion of testing and security validation before production release.*

- Risk Management

#	Risk Description	Type	Severity	Solution
1	The AI system may incorrectly extract or structure knowledge components from uploaded materials	Technical Risk	High	Enable manual correction of knowledge components, perform regular testing, and adjust the AI logic as necessary
2	Uploading large PDF/PPT files may slow processing or affect system responsiveness	Performance Risk	Medium	Implement file size limits, optimize parsing processes, and use asynchronous background processing
3	Weak authentication or data handling may expose user data	Security Risk	High	Use secure authentication (Firebase), encrypt sensitive data
4	Errors in the Bayesian Knowledge Tracing (BKT) implementation may result in inaccurate mastery probability updates	Technical Risk	High	Perform early prototyping, unit testing of the BKT algorithm, and validate results using sample datasets before deployment

Testing Plan

Testing Level	Focus Area
Unit Testing	Individual functions (BKT math, text extraction).
Integration Testing	Communication between React, FastAPI, and Firebase.
System Testing	The complete user journey from PDF upload to Quiz results.
User Acceptance (UAT)	Validating that the tool solves the "reading-only" problem.